Python Basics Assignment

Question 1: What is Python and why is it widely used in Data Analytics? Mention any three reasons.

Answer:

Python is a high-level, open-source programming language that is simple to read and write. It was created by Guido van Rossum in 1991 and has become the most popular language in the world for Data Analytics, AI and Machine Learning.

Reason 1: Simple and Easy to Learn:

- Python's syntax is very clean and readable it almost reads like plain English
- A beginner can start writing useful code much faster compared to languages like Java or C++
- Less time learning the language = More time actually analyzing data

Reason 2: Powerful Libraries for Data Analysis:

- Python has ready-made tools called **libraries** built specifically for data work
- **Pandas** → for data cleaning and manipulation
- **Matplotlib / Seaborn** → for data visualization
- **NumPy** → for numerical calculations
- You don't have to build everything from scratch these libraries do the heavy lifting

Reason 3: Free and Has a Huge Community:

- Python is completely **free and open source** no license cost unlike tools like MATLAB
- Millions of users worldwide mean **tons of tutorials, solutions and support** available online
- Any error or problem you face someone has already solved it and posted the answer

Question 2: Explain the difference between List and Tuple in Python.

Answer:

A List is a collection of items that is **changeable (mutable)** — meaning you can add, remove or modify items after creating it.

my_list = [1, 2, 3, 4, 5]

- Uses **square brackets [ ]**
- Items **can be changed** after creation

What is a Tuple?

A Tuple is a collection of items that is **unchangeable (immutable)** — meaning once created, you **cannot modify** it.

```
my_tuple = (1, 2, 3, 4, 5)
```

- Uses **round brackets ( )**
- Items **cannot be changed** after creation

Question 3: What is a function in Python? Why are functions useful?

Answer:

A function is a block of reusable code that performs a specific task. You define it once and can call/use it as many times as you want anywhere in your program.

```
def greet():
    print("Hello, Welcome to Python!")
greet()    # Calling the function
greet()    # Calling it again
```

- **def** keyword is used to define a function
- **greet()** is the function name
- The code inside runs every time you call it

Types of Functions in Python

1. Built-in Functions *(already available in Python)*

```
python
print()     # prints output
len()       # gives length
input()     # takes user input
```

2 . User Defined Functions *(you create them yourself)*

```
python
def add(a, b):
    print(a + b)
add(5, 3)    # Output: 8
```

Why are Functions Useful?

Reason 1 Reusability:

- Write the code once, use it multiple times
- No need to write the same code again and again

python

```
# Without function repeating code
print(5 + 3)
print(10 + 3)
print(15 + 3)

# With function clean and reusable
def add(a, b):
    print(a + b)
add(5, 3)
add(10, 3)
add(15, 3)
```

Reason 2 Organized and Readable Code:

- Large programs become easier to read and manage
- Each function handles **one specific task** — keeps things clean

Reason 3 Easy to Fix/Update:

- If something needs to change, you update it in **one place only**
- Without functions you'd have to find and fix the same code in 10 different places

Question 4: Write a Python program to take a user's name as input and print a greeting message.

Answer:

The code is as follows:

```
name = input("Enter your name: ")
print("Hello,", name, "! Welcome to Python!")
```

The Example Output:

Enter your name: Rahul

Hello, Rahul ! Welcome to Python!

Question 5: Write a Python program to check whether a number is even or odd using conditional statements.

Answer:

The code is as follows:

```
number = int (input ("Enter a number: "))
if number % 2 == 0:
    print (number, "is Even")
else:
    print (number, "is Odd")
```

The Example Output:

Enter a number: 8

8 is Even

Enter a number: 7

7 is Odd

Question 6: Write a program to print numbers from 1 to 10 using a loop.

Answer:

The Code is as follows:

```
i = 1
while i <= 10:
    print(i)
    i = i + 1
```

The Example Output:

1

2

3

4

5

6

7

8

9

10

Question 7: Create a list of five numbers and print the maximum number from the list.

Answer:

The Code is as follows:

```
numbers = [12, 45, 7, 89, 34]
maximum = numbers[0]
for i in numbers:
    if i > maximum:
        maximum = i
print("List of numbers:", numbers)
print("Maximum number:", maximum)
```

The Example Output:

List of numbers: [12, 45, 7, 89, 34]

Maximum number: 89

Question 8: Write a Python program to remove duplicate values from a list using a set.

Answer:

The Code is as follows:

```
numbers = [1, 2, 3, 2, 4, 3, 5, 1, 6]
unique_numbers = []
for i in numbers:
    if i not in unique_numbers:
        unique_numbers.append(i)
print("Original list:", numbers)
print("List after removing duplicates:", unique_numbers)
```

The Example Output:

```
Original list: [1, 2, 3, 2, 4, 3, 5, 1, 6]
List after removing duplicates: [1, 2, 3, 4, 5, 6]
```

Question 9: Write a function that returns the square of a number.

Answer:

The code is as follows:

```
def square(number):
    return number ** 2
number = int(input("Enter a number: "))
print("Square of", number, "is:", square(number))
```

The Example Output:

```
Enter a number: 5
Square of 5 is: 25
```

Question 10: Write a Python program to count how many times a number appears in a list. Example List [2,3,4,2,5,2]

Answer:

The Code is as follows:

```
numbers = [2, 3, 4, 2, 5, 2]

unique_numbers = set(numbers)

for num in unique_numbers:

    print(num, "appears", numbers.count(num), "times")
```

The Example Output:

2 appears 3 times

3 appears 1 times

4 appears 1 times

5 appears 1 times